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A Decision Support System for Business Continuity and Critical Service Recovery Prioritization

A Master's Project Submitted in Partial Fulfillment of the Requirements for the Master's Degree in Information Systems

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ABSTRACT

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TABLE OF CONTENTS

Abstract 2

الخلاصة 3

Table of Contents 4

List of Abbreviations 6

List of Figures 7

List of Tables 8

Acknowledgements 9

Chapter 1: Introduction 10

1.1 Motivation and Scope 10

1.2 Problem Statement and Significance for Information Systems 11

1.3 Importance of the Problem and the Big Picture 13

1.4 Objectives and Proposed Solution 14

Chapter 2: Literature Review 16

2.1 Business Continuity Management 17
- 2.1.1 Continuity, Recovery, and Organizational Resilience 17
- 2.1.2 International Standards and Guidance 18
- 2.1.3 Saudi National and Digital-Government Guidance 20

2.2 Business Impact Analysis and Recovery Requirements 21
- 2.2.1 Units of Analysis and Impact Evidence 21
- 2.2.2 Representing Impact over Time 23
- 2.2.3 MTPD, RTO, and Related Parameters 24
- 2.2.4 Comparative Evidence from BIA Studies 25

2.3 Service Criticality and Recovery Prioritization 27
- 2.3.1 Multi-Criteria Approaches 27
- 2.3.2 Configurable Rule-Based Approaches 28
- 2.3.3 Priority Rankings and Recovery Schedules 29

2.4 Dependency Modeling and Supporting-Resource Criticality 30
- 2.4.1 Service and Resource Relationships 30
- 2.4.2 Interdependencies and Impact Propagation 31
- 2.4.3 Assigning Criticality to Shared Dependencies 33

2.5 Decision Support Systems and Comparable Applications 34
- 2.5.1 Decision Support Foundations 34
- 2.5.2 Research Prototypes and Analytical Applications 35
- 2.5.3 Continuity Management Software 35

2.6 Research Methods and Evaluation Evidence 38
- 2.6.1 Design Science and Artifact Development 38
- 2.6.2 Correctness, Usefulness, and Operational Effectiveness 39
- 2.6.3 Transferability and Evidence Quality 40

2.7 Comparative Synthesis and Research Gap 41
2.8 Chapter Summary 43
Chapter 3: Methodology/ Research Design 44
Chapter 4: Results and Discussions 45
Chapter 5: Analysis/ Discussions 46
Chapter 6: Conclusions 47
References 48
Appendices 51

LIST OF ABBREVIATIONS

If appropriate, list any abbreviations in alphabetic order used in the thesis.

CDF	Cumulative Distribution Function
KSU	King Saud University

LIST OF FIGURES

No table of figures entries found.

LIST OF TABLES

Table 2.1: Standards and guidance documents compared by scope, treatment of business impact analysis, and nature of the requirements. 19

Table 2.2: Related analytical studies: decision object, approach, and evaluation limits. 26

Table 2.3: Comparable continuity tools: documented functionality and limits of the reviewed evidence. 36

Table 2.4: Synthesis of prior contributions and implications for TDRP and Masar....41

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Chapter 1: Introduction

This chapter establishes the context, research problem, and objectives of the proposed decision support system for business continuity and critical service recovery prioritization. Section 1.1 presents the motivation and scope. Section 1.2 defines the research problem and its significance for information systems. Section 1.3 situates the problem within the broader organizational continuity context, and Section 1.4 sets out the objectives, the proposed TDRP methodology, and its implementation in Masar.

1.1 Motivation and Scope

Organizations are required to keep delivering their essential services even when operations are disrupted, whether the disruption reaches their activities, infrastructure, information, or the external parties they depend on. How serious an interruption turns out to be depends on more than the incident itself. It also depends on the importance of the affected service and on how long that service stays unavailable. An interruption that can be absorbed at first may become unacceptable once its effects on service users, operations, financial obligations, regulatory requirements, and institutional reputation begin to accumulate. For this reason, business continuity management calls on organizations to identify their priority services, to understand how the impacts of an interruption grow over time, and to set recovery requirements they can defend before any disruption takes place [1], [2].

Business continuity management (BCM) offers a structured way to prepare for disruptions and to maintain or restore prioritized activities and services. Within it, business impact analysis (BIA) plays a central part: it studies the consequences of an interruption, establishes how important services and activities are relative to one another, and feeds into recovery planning. Two recovery parameters matter most for this research. The maximum tolerable period of disruption (MTPD) is the duration beyond which the consequences of an interruption become unacceptable, and the recovery time objective (RTO) is the target time for resuming a service or activity. The two are related but serve different purposes, and each has to be set with reference to

the impacts the organization has assessed and the recovery requirements it has identified [1], [2].

Assessing the consequences of an interruption is inherently a matter of time. Services differ in how quickly the impacts of an interruption become serious, and organizations may also differ in the criteria they use to judge those impacts. The business continuity guidance of the Digital Government Authority (DGA), for instance, presents impact-based classifications and recovery targets for government digital platforms, and at the same time recognizes that the time intervals used for assessment should follow the sensitivity of the activities concerned [2]. These principles point toward an approach in which each organization defines its own assessment periods, impact criteria, acceptable thresholds, and matching recovery rules, rather than assuming that one fixed configuration suits everyone.

This project proposes Time-Dependent Recovery Prioritization (TDRP), a configurable rule-based methodology, and Masar, a web-based decision support system (DSS) that implements it. The approach starts from business services and their BIA results, applies explicit rules to their time-dependent impact assessments, and derives criticality, recovery requirements, and relative recovery priorities. The dependencies that support each service are then related to its criticality, and the system presents the outcome on an operational dashboard. Its purpose is to help responsible personnel interpret and use continuity information, not to replace their professional judgment or to take on responsibility for operational recovery decisions.

The scope of the research covers the design, implementation, and evaluation of the TDRP methodology within Masar. The central concern is the service-level assessment and prioritization process. Supporting dependencies, that is, the elements and resources on which service delivery relies, are considered only to the extent that their relationships help explain how services are delivered and what they require for recovery.

1.2 Problem Statement and Significance for Information Systems

Deciding on recovery priorities takes more than recognizing that a service is important. An organization has to be able to explain how the assessed impacts of an

interruption justify a service's criticality, how those impacts relate to acceptable disruption durations and recovery targets, and how the resulting priorities should be read alongside the dependencies that the service needs in order to be delivered. National business continuity guidance names BIA, recovery requirements, and dependencies as important components of continuity management [1], [2]. Yet these principles do not lay down a single mandatory computational procedure that every organization must follow to turn successive impact assessments into an MTPD, an RTO, and service recovery priorities. Nor can the thresholds or assessment intervals of one organization be taken as suitable for any other.

The research problem is therefore the need for a clear, configurable, and reproducible method that translates time-dependent impact assessments of business services into recovery requirements and relative recovery priorities, while keeping an explicit connection between the assessed services and the dependencies that support their delivery. Such a method should make its decision rules visible and let different organizations configure them according to their approved criteria. Its results should also be traceable to the assessments underlying them. In addition, it should distinguish the priority assigned to a service from the importance given to a supporting dependency because other services rely on it. The project does not assume that continuity practices are absent today or that organizations lack information. It addresses, instead, the formalization and implementation of one particular service-first prioritization procedure.

The problem matters to information systems because it concerns the transformation of organizational assessments into consistent and explainable decision-support outputs. A DSS can represent services, time-based impacts, recovery parameters, configurable rules, and dependency relationships in a single coherent information model, and it can apply the same declared logic to comparable inputs. This makes it possible to inspect how a given priority was obtained, to examine what happens when an approved rule configuration is changed, and to present the resulting information to continuity practitioners. Designing and evaluating a purposeful artifact of this kind is in line with design-science research, which is concerned with building and assessing artifacts that address identified organizational problems [3], [4].

1.3 Importance of the Problem and the Big Picture

Business continuity is an organizational capability and not merely a technical recovery activity. A disruption can affect the availability of public or private services, the obligations an organization must meet, the experience of its beneficiaries, and its capacity to keep essential operations running. The Saudi National Framework for Risk, Emergency and Business Continuity Management treats business continuity as one part of a broader framework, relates it to the ability of an organization to withstand disruption, and calls for its alignment with the organization's own strategy [1]. The DGA likewise stresses sustained service delivery and the alignment of recovery arrangements with organizational objectives [2]. Seen from this angle, prioritization is important because organizations need to determine which service outcomes are the most time-sensitive and to let those priorities inform their recovery efforts.

A service-oriented view also offers a principled way of treating supporting dependencies. Services are delivered through interacting activities and enabling resources, and the interruption of a shared dependency can affect more than one service. The criticality of a supporting dependency should therefore be read in relation to the services it enables, and not inferred from its own characteristics alone. In the proposed methodology, service criticality is established first. The relevant dependency relationships are recorded next, and a stated rule links each supporting dependency to the criticality of the service level. Where several services share a dependency, the methodology has to define how their classifications are combined. Taking the highest criticality among the supported services is a rule proposed here for examination and validation. It is not a general requirement that the reference frameworks can be said to impose.

Another concern is how to balance standardization against organizational flexibility. Consistency calls for a common method of applying rules, whereas practical use calls for organizations to choose impact scales, assessment time frames, and recovery thresholds that fit their own contexts. A configurable rule-based methodology can meet both needs by standardizing the form and execution of the decision logic without requiring identical parameter values from every organization. The contribution anticipated in this sense is not a replacement for existing BCM standards or for the

expertise of practitioners. It is a transparent procedure for interpreting BIA results and expressing recovery priorities. Research on business impact analysis and on integrated continuity planning gives a broader basis for studying the link between organizational impacts and recovery planning [5], [6].

1.4 Objectives and Proposed Solution

The aim of this project is to design, implement, and evaluate a configurable rule-based methodology that derives recovery requirements and prioritizes critical business services from time-dependent impact assessments. The methodology, TDRP, is the main research contribution. Masar is the proposed web-based DSS in which the methodology will be configured, applied, and evaluated, and through which its results will be shown to authorized users. More specifically, the project pursues five objectives. The first is to derive service criticality, MTPD, and RTO from time-dependent impact assessments through configurable rules, and the second is to establish relative recovery priorities from criticality and recovery time requirements. The third is to associate services with their supporting dependencies and to assign criticality to those dependencies by a documented rule. The fourth is to implement the methodology in Masar, with an operational dashboard that presents its results, and the fifth is to evaluate the correctness, consistency, and traceability of the methodology and its implementation.

The methodology will represent BIA findings at the service level across successive assessment periods and the relevant impact dimensions. Following the criteria and decision rules defined by the organization, it will determine service criticality, identify the applicable MTPD, and establish the corresponding RTO. It will also document how temporal assessments, impact thresholds, and recovery-time mappings are interpreted, including how incomplete data are handled and what happens when an assessed impact does not reach a threshold. Organizations will be able to configure the permitted parameters so that they reflect their approved continuity requirements.

From these service-level results, TDRP will set relative recovery priorities through explicit rules that take criticality and recovery-time requirements into account. The approach will then associate each assessed service with the dependencies needed to deliver it. A supporting dependency will inherit criticality from the service or services it

supports according to a documented rule. For a dependency shared by several services, the proposed default is the highest criticality among them, subject to evaluation. The purpose of this association is to preserve the business reasoning behind the importance of supporting dependencies. A ranking of services by priority is not to be equated with an automatically executable sequence of recovery operations.

Masar will put the methodology into practice through an integrated data model and rule processing functions. Authorized users will be able to maintain services, impact assessments, configuration settings, recovery requirements, and the relevant dependency relationships. An operational dashboard will present the resulting service classifications, MTPD and RTO values, recovery priorities, and linked supporting dependencies. During an actual interruption, users can consult the recorded information on the affected services to see their established priority and dependencies. The dashboard will show the outputs of the methodology together with their context. It will not simulate incidents, infer operational actions, recommend recovery procedures on its own, or make or carry out recovery decisions.

The research will evaluate the methodological rules and their implementation. The evaluation will look at the consistency and reproducibility of the outputs, the behavior of the method under different valid configurations, the correctness of the derived MTPD and RTO values against specified test cases, the treatment of shared dependencies, and the traceability of displayed priorities to the impact assessments and rules they come from. The intended deliverables are a documented TDRP methodology, an implemented Masar prototype, and an evaluation of their correctness and usefulness as a basis for structured decision support. How effective the approach is in real incidents lies beyond what a prototype-level evaluation alone can establish.

Chapter 2: Literature Review

Business continuity decisions connect organizational obligations with the time available to sustain or restore essential services. This chapter reviews the concepts, analytical methods, and information systems that support those decisions. Its central concern is the transformation of business impact analysis (BIA) into defensible recovery requirements and relative service priorities, together with the interpretation of supporting dependencies. The review provides the basis for assessing the contribution of Time-Dependent Recovery Prioritization (TDRP) and its implementation in Masar, within the scope established in Chapter 1.

Section 2.1 establishes the management context and relevant guidance. Section 2.2 examines impact assessment and recovery parameters. Section 2.3 compares prioritization and scheduling approaches, and Section 2.4 examines dependency models. Section 2.5 reviews decision support and comparable software. Section 2.6 evaluates the research methods used in related work. Section 2.7 synthesizes the evidence and positions the research contribution, followed by the chapter summary in Section 2.8.

The review is a focused thematic synthesis rather than an exhaustive systematic review. Sources were selected for their relevance to continuity management, temporal BIA, service or asset criticality, dependencies, and decision-support evaluation. The evidence comprises research articles, government guidance, public standards descriptions, and product documentation. Foundational contributions are considered alongside recent applications; methodological relevance, rather than publication date alone, determines inclusion. Research findings are distinguished from normative guidance and advertised capabilities. Public ISO catalogue records support statements about scope, but not clause-level interpretations of the full standards. Product documentation establishes disclosed functions rather than independently measured effectiveness. These distinctions constrain the conclusions drawn from the review.

2.1 Business Continuity Management

2.1.1 Continuity, Recovery, and Organizational Resilience

Business continuity concerns the sustained delivery of prioritized products and services at an acceptable capacity when normal operations are disrupted. BCM organizes the responsibilities, assessments, plans, exercises, and improvement activities through which that capability is maintained. A business continuity management system (BCMS) formalizes these activities within an organizational management system. The scope of ISO 22301 places continuity within such a system rather than treating it as an isolated technical plan [7]. This distinction is consequential for information systems: an application may support selected BCM activities without constituting the organization's entire BCMS.

Continuity and disaster recovery overlap, but their decision objects differ. Continuity planning is concerned with the delivery of required business outcomes, potentially through temporary arrangements or reduced operating levels. Information-system recovery concerns the restoration of the technical resources that enable those outcomes. NIST SP 800-34 distinguishes information-system contingency planning from business continuity, crisis communications, incident response, and other related plans [8]. A successfully restored server therefore does not, by itself, demonstrate that a business service has resumed at its required capacity. Personnel, data reconciliation, facilities, suppliers, and procedural prerequisites may remain unavailable.

Organizational resilience supplies a wider context. Sahebjamnia et al. connect continuity and recovery planning through the loss of operating capability over time and the resources needed to regain it [6]. Their analysis shows why short-term continuity and longer-term recovery should not be treated as independent demands on unlimited resources. However, resilience encompasses more than recovery speed. The asset-criticality framework of Aghabegloo et al., for example, considers absorptive, adaptive, and restorative capacities alongside sustainability criteria [9]. These broader perspectives help explain the significance of service recovery without implying that a prioritization method measures resilience in its entirety.

Risk assessment and BIA also answer related but different questions. Risk assessment examines uncertain events and their consequences; BIA establishes the

significance of disrupted activities or assets in relation to organizational objectives. NIST IR 8286D connects BIA results to enterprise risk prioritization, distinguishing availability-related criticality from sensitivity associated with confidentiality and integrity [10]. A service can have severe consequences if unavailable even when the probability of a particular initiating event is low. Combining likelihood and impact into a risk score should not silently replace the analysis of tolerable disruption duration.

2.1.2 International Standards and Guidance

ISO 22301:2019 identifies requirements for establishing and maintaining a BCMS, while ISO 22313:2020 provides guidance on applying those requirements. ISO/TS 22317:2021 focuses on a documented BIA adapted to organizational needs, objectives, resources, and constraints [7], [11], [12]. The public description of ISO/TS 22317 explicitly allows different BIA processes. This adaptability supports contextual assessment, but it should not be interpreted as evidence that all computational procedures are equally appropriate. Organizations still need to justify the relationship between assessment criteria, tolerability, and recovery requirements.

Table 2.1: Standards and guidance documents compared by scope, treatment of business impact analysis, and nature of the requirements.

Document	Scope	Business impact analysis and recovery parameters	Nature of the text
ISO 22301:2019 [7]	Establishing and maintaining a business continuity management system suited to the organizational context.	Not stated in the published scope description.	Requirements standard.
ISO 22313:2020 [11]	Guidance on applying ISO 22301 to organizations of different types and contexts, so that delivery is sustained at a predefined acceptable capacity.	Not stated in the published scope description.	Guidance.
ISO/TS 22317:2021 [12]	Guidance on conducting a documented business impact analysis.	Adapts the analysis to organizational needs, objectives, resources, and constraints, and explicitly does not impose a single uniform process.	Technical specification (guidance).
NIST SP 800-34 Rev. 1 [8]	Contingency planning for federal information systems; comprehensive business-process continuity is outside its scope.	Three analysis tasks (process criticality, resources, recovery priorities). Defines MTD, RTO, and RPO, and states that the RTO should normally be shorter than the MTD. Times and costs in the guide are illustrative.	Guidance with templates and examples.

Note: The entries for the ISO documents are based on their public catalogue descriptions.

NIST SP 800-34 provides a more operational account for federal information systems. Its BIA identifies process criticality, supporting resources, and resource recovery priorities. Maximum tolerable downtime (MTD) expresses the outage a business process can accept, whereas RTO expresses the recovery target for a supporting system resource. RTO should normally be shorter than MTD because resumption may also require work recovery after technical restoration. RPO concerns the acceptable point to which data can be recovered; it is not an additional forward-

running component of the outage budget [8]. These definitions prevent a common conceptual error: treating every recovery-related time quantity as interchangeable.

The distinction between guidance and algorithm is equally important. NIST supplies a process, terminology, and examples through which an organization can establish requirements. Its illustrative durations do not constitute universal thresholds. Similarly, public ISO scope descriptions cannot establish that the complete standards either prescribe or omit a particular formula. The justified conclusion is narrower: the reviewed guidance supports organization-specific BIA and the explicit identification of recovery requirements, while the detailed computation used by a DSS remains an object requiring its own specification and assessment.

Research on BCM frameworks reinforces this need for interpretation. Russo et al. compare continuity-related frameworks and discuss the difficulty of operationalizing organizational maturity assessment [13]. Arias-Aranda et al. identify a reference set of BCMS processes through standards analysis and practitioner verification [14]. Such studies clarify the management activities within which a BIA tool operates. They do not demonstrate the correctness of a particular impact-to-RTO calculation. Agreement that BIA belongs in a BCMS is evidence about process relevance, not validation of every method implementing that process.

2.1.3 Saudi National and Digital-Government Guidance

The Saudi National Framework for Risk, Emergency and Business Continuity Management situates continuity within a broader organizational capability for managing disruption. Its BCM material connects essential services and products with enabling processes, impact assessment, recovery requirements, resources, and dependencies [1]. This service-oriented structure is directly relevant to the thesis: technical-resource importance is interpreted in relation to the organizational outcomes that resources enable.

The framework allows data collection through interviews, workshops, and questionnaires. Impact categories extend beyond direct financial loss to legal, operational, reputational, beneficiary, health and safety, and contractual consequences. Assessment times and classifications are adapted to the organization. Its example

recovery classes illustrate how requirements may be expressed; they should not be adopted as universally mandatory cutoffs [1]. A configurable system should consequently separate the structure of an assessment from the particular parameter values approved by an organization.

The DGA guidance similarly relates BIA to an approved impact matrix, time-dependent disruption assessment, and the recovery requirements of digital government. Its outputs include MTPD, RTO, RPO, minimum business continuity objective (MBCO), and necessary resources [2], [15]. The assessment horizon depends on service sensitivity. The English issue also presents platform-impact tiers with associated target recovery times and adopts the highest level where impact factors produce different classifications [15]. This is substantive prior art for explicit classification and mapping rules, even though it does not by itself settle every temporal interpretation needed by TDRP.

The national documents establish both a contextual basis and a constraint on the proposed contribution. Masar should accommodate approved organizational scales, but configurability is not permission to select arbitrary values without governance. A threshold represents a judgment about unacceptable consequences, and an RTO represents a target with operational implications. Their approval, ownership, and revision remain organizational responsibilities. A DSS can preserve and execute these decisions consistently; it cannot establish their appropriateness merely by producing a numerical result.

2.2 Business Impact Analysis and Recovery Requirements

2.2.1 Units of Analysis and Impact Evidence

The first methodological choice in BIA is the object being assessed. Studies variously examine products, business processes, services, datasets, systems, and physical assets. Horalek investigates automated meter management (AMM) data and supporting systems, while Budiyanto et al. examine the consequences of electricity disruption at a container terminal [16], [17]. NIST starts from organizational processes and connects them to enabling resources [8], [10]. These objects are related, but a score

assigned to a dataset cannot be assumed to have the same interpretation as a score assigned to an end-to-end public service.

A service-level assessment is useful when the decision concerns the restoration of outcomes delivered to beneficiaries. It can preserve the business rationale for subsequently identifying important resources. However, a service may contain activities with different deadlines, operate at several locations, or depend on alternative delivery channels. Defining service boundaries too broadly can conceal urgent components; defining them too narrowly can produce duplicate assessments and obscure shared obligations. The unit of analysis must therefore be stated before comparing criticality or recovery times across services.

Much BIA evidence is elicited from knowledgeable participants. Horalek combines technical analysis with guided interviews involving service sponsors and dataset owners [16]. Budiyanto et al. use management questionnaires and a focus group to establish relevant impacts and assessment scales, supplementing these judgments with operating records [17]. Interviews can identify consequences that do not appear in transaction logs, including reputational damage and obligations to beneficiaries. Their limitations include inconsistent interpretations, incomplete knowledge of dependencies, and differences in how departments understand severe impact.

Operational data can strengthen such assessments without removing judgment. The container-terminal study relates disruption to productivity, throughput, working time, economic loss, and other consequences [17]. Historical records help anchor the magnitude of observed effects, but a single outage does not reveal every future impact trajectory. A disruption at peak demand may differ from one during a quiet period, and a longer interruption can activate penalties or dependencies absent from the observed event. Data-supported BIA remains a combination of evidence, assumptions, and organizational interpretation.

Impact categories also require careful separation. Financial loss is often quantifiable, whereas reputation, beneficiary harm, legal exposure, and service quality may require ordinal judgments. NIST's distinction between criticality and sensitivity further shows why availability, confidentiality, and integrity should not be collapsed without an explicit purpose [10]. An availability-focused recovery method may

legitimately use several consequence categories, but it should not reinterpret all information-security assessments as downtime urgency. The source and meaning of each category should remain visible in the resulting decision record.

2.2.2 Representing Impact over Time

A static impact label records severity without showing how quickly it arises. Temporal BIA instead evaluates interruption consequences at successive durations. The AMM study uses seven availability-assessment horizons, ranging from thirty minutes to more than one week [16]. This approach preserves an important difference between services that eventually experience similarly severe consequences but reach them at different times. For recovery prioritization, the onset of unacceptable impact may matter as much as the highest impact reached within the assessment horizon.

A time-indexed matrix is a practical representation of this evidence. Its rows or columns identify impact dimensions and assessment durations, and its cells record the corresponding ratings. The matrix remains an elicited or observed assessment, not a continuous physical law. A rating at eight hours does not reveal exactly what happened at every preceding minute. If an unacceptable rating is first observed at eight hours after an acceptable rating at four hours, locating the transition within that interval requires additional evidence or an explicit convention. Assigning eight hours as a policy boundary can be reproducible, but it should not be described as a measured crossing time.

Temporal resolution therefore creates a methodological trade-off. Closely spaced intervals can improve the usefulness of a matrix for sensitive services, but require more assessment effort and can suggest unjustified precision. Coarse intervals reduce elicitation demands while making threshold interpretation less exact. The DGA guidance recognizes variation in appropriate time scales across activities [15]. A configurable method should preserve the declared horizon and time units, since identical ordinal ratings on different time grids do not necessarily represent equivalent urgency.

Time-dependent impact need not be uniformly increasing in every category. Cumulative financial loss may increase with duration, but deadline-based consequences can change abruptly, and temporary workarounds can reduce particular operational effects. The quantitative threat model QuantTM represents impact through recovery

stages and time frames rather than a single static loss [18]. Although its expected-loss purpose differs from service prioritization, it demonstrates why assumptions about persistence and recovery levels affect temporal interpretation. A method that requires monotone ratings must declare that requirement and identify inconsistent input rather than silently rearranging it.

Missing assessments create a separate problem. An empty cell may indicate insufficient information, an inapplicable category, or an assessment that has not yet been completed. None is necessarily equivalent to negligible impact. Likewise, failure to reach an unacceptable level within the observed horizon does not establish unlimited tolerability. These are logical consequences of working with incomplete, bounded assessments. They motivate explicit output states and explanations in a computational implementation, rather than the use of a convenient numerical default that could be mistaken for a confirmed recovery requirement.

2.2.3 MTPD, RTO, and Related Parameters

MTPD describes the tolerability boundary for disruption, while RTO expresses the target for resumption. The relationship between them depends on the assessed object and the activities required to restore acceptable service. NIST's distinction between process MTD and resource RTO makes this explicit: time may still be needed after system restoration to recover work and resume the process [8]. Saudi guidance uses MTPD alongside RTO in the continuity assessment [1], [15]. The terms can be compared conceptually, but calculations must preserve their original unit of analysis.

An RTO is therefore neither an observed recovery duration nor a guarantee of technical capability. It expresses a requirement or approved target that should inform recovery arrangements. A recovery exercise may demonstrate that existing arrangements fail to meet it. Conversely, a rapidly recoverable system may support a service whose tolerance is less demanding. A DSS should distinguish the target from demonstrated capability so that infeasible requirements remain visible as planning issues instead of being overwritten by current technical limitations.

MBCO adds a capacity dimension. Resuming a service at an acceptable reduced level is different from restoring its normal throughput. Sahebjamnia et al. represent

operating levels and the requirements for maintaining or restoring them under resource constraints [6]. Their work illustrates why a time target without a specified operating condition can be ambiguous. For TDRP, the reviewed literature supports treating recovery requirements as contextual decisions: the relevant time, assessed service, and meaning of resumption should be recorded together even where capacity optimization lies outside the prototype.

RPO concerns tolerable data loss measured by reference to the point to which data must be recoverable [8]. It should not be inferred solely from service downtime or treated as a second name for RTO. Terminology also needs checking when studies are compared. The planning model of Sahebjamnia et al. uses RPO in connection with a resumption operating level [6]; that usage is not adopted here. This difference is a reminder that shared abbreviations do not guarantee shared constructs across continuity research.

The AMM study relates time-based impacts to tolerable periods and proposes recovery objectives [16]. It is a close methodological precedent because the analysis links availability consequences across durations with recovery requirements. Its findings nevertheless remain grounded in one technical domain and its selected assessment criteria. A general-purpose service method requires a more explicit separation between the configurable policy, the calculation that executes it, and the evidence supporting the original ratings. That separation enables reproducibility without claiming that the ratings themselves are objective measurements.

2.2.4 Comparative Evidence from BIA Studies

The analytical studies can be distinguished by what they produce. Matrix-based BIA supports impact classes and recovery requirements. Data-assisted BIA relates assessments to operating records. Multi-criteria models produce relative criticality scores; quantitative threat models estimate loss; process-mining models propagate relative incident effects; and optimization models allocate resources or determine recovery trajectories. Table 2.2 compares representative studies. Their outputs should not be treated as interchangeable measures of recovery priority.

Table 2.2: Related analytical studies: decision object, approach, and evaluation limits.

Study	Inputs, method, and outputs	Temporal treatment and evidence
Horalek [16]	AMM datasets and systems; technical analysis and guided interviews; impact matrices inform tolerable periods and recovery objectives.	Availability assessed at seven durations. One domain-specific worked case; no controlled comparison of incident recovery outcomes.
Budiyanto et al. [17]	Container-terminal electricity disruption; management judgments and operating records inform five-level impact classifications.	Observed disruption and subsequent consequences. Supports contextual impact assessment rather than a general MTPD or RTO derivation rule.
Sahebjamnia et al. [6]	BIA requirements, importance weights, resources, and budgets enter a multi-objective planning model; outputs include operating trajectories and allocations.	Discrete recovery periods. Manufacturing case and hypothetical disruption evaluation; tolerability requirements are inputs to planning.
Aghabegloo et al. [9]	Expert judgments, historical data, and risk scenarios; fuzzy decision methods and a Bayesian network produce asset criticality and contributor rankings.	Recovery-related quantities enter resilience assessment. Gas-plant case, expert face validation, and sensitivity analysis; no generic temporal service-threshold rule.
Von der Assen et al. [18]	Threat probabilities and business-loss assumptions enter QuantTM; a browser prototype estimates expected financial loss.	Loss represented through recovery stages and time frames. Preprint and case-based evaluation; expected loss is not service recovery priority.
Raptaki et al. [19]	Procurement event logs generate a weighted process graph; incident impact is propagated to rank activities.	Scenario-sensitive relative impact rather than tolerable-duration derivation. One organizational case with comparisons and interviews; coverage depends on logged activities.

Horalek's study is valuable for its explicit temporal matrix, but its evaluation is principally a worked assessment of an AMM environment [16]. Budiyanto et al. provide operational grounding for impact classification in a container-terminal disruption, yet the study does not establish a general impact-to-MTPD algorithm [17]. Both show the importance of domain knowledge. Their contribution to a transferable DSS lies in the

structure of assessment and the evidence-collection process, rather than in adopting their numerical scales without further justification.

Aghabegloo et al. extend the analytical scope by linking key products with physical-asset criticality under sustainability and resilience considerations [9]. Their method combines expert judgments and quantitative inputs through several decision techniques. Raptaki et al. instead derive a process graph from event logs and propagate incident-related impact across activities [19]. These studies demonstrate that richer models can expose relationships missed by a simple matrix, while also increasing the requirements for elicitation, data preparation, and model interpretation.

2.3 Service Criticality and Recovery Prioritization

2.3.1 Multi-Criteria Approaches

Criticality assessment usually involves several consequences that cannot be expressed in one natural unit. Multi-criteria decision-making methods offer an explicit means of combining these judgments. Aghabegloo et al. use fuzzy best–worst weighting and fuzzy TOPSIS to identify key products, followed by fuzzy DEMATEL and a Bayesian network to evaluate supporting physical assets [9]. The approach recognizes that economic, social, environmental, and resilience factors may influence one another rather than contributing as isolated criteria.

The gas-plant application illustrates both the power and demands of this approach. Product selection precedes asset assessment, retaining a business rationale for the asset priorities. Expert judgments, historical information, and risk scenarios inform the model. Its asset criticality index combines weighted probabilities associated with critical factors, and further analysis identifies influential contributors [9]. The result provides a basis for directing attention and continuity investment. It does not directly supply a service's tolerable interruption duration from successive ordinal ratings.

Weighted aggregation requires a defensible interpretation of the scales and trade-offs. An average can allow low values in several dimensions to offset an unacceptable value in another. That may be appropriate when criteria are compensatory, but inappropriate when a single legal or safety consequence is decisive. Fuzzy methods represent imprecision in judgments; they do not eliminate the need to justify which

judgments can compensate for others. Likewise, a probabilistic index is meaningful only in relation to its variables, elicitation assumptions, and network structure.

The criteria dependencies in Aghabegloo et al. also differ from service-resource dependencies. Their Bayesian network represents relationships among factors affecting asset criticality [9]. It should not be described as equivalent to recording which services use a shared database. The former models how criteria interact in assessment; the latter models operational support relationships. Confusing these objects would obscure both the strengths of the multi-criteria approach and the narrower purpose of Masar's dependency map.

Multi-criteria methods are therefore relevant alternatives when decision makers need explicit trade-offs across several dimensions and can support the necessary elicitation. For TDRP, their principal lesson is that aggregation semantics must be transparent. A simpler rule is not inherently more valid, but may be more readily inspected when the intended policy is a categorical tolerability threshold rather than a compensatory utility score. Comparative evaluation should ask whether the method represents the organization's decision policy correctly, rather than assuming that greater mathematical complexity implies better prioritization.

2.3.2 Configurable Rule-Based Approaches

Rule-based approaches apply declared conditions and mappings to assessed impacts. A rule may select a criticality category, identify a tolerability boundary, or map an assessed class to an approved recovery target. The DGA's highest-impact classification is an example of a non-compensatory policy: a severe consequence is not diluted by lower ratings elsewhere [15]. This approach is intelligible to practitioners because the decisive category can be identified, but its appropriateness still depends on the definition of the impact scale and the intended use of the result.

Commercial documentation provides closer precedents for temporal rules. GlobalSuite records impacts over configurable time scales and documents automatic MTPD criteria based on a selected impact level, an average, or a specific impact type. It also supports consolidation across individual assessments through average, weighted-average, maximum, or minimum rules [20]. ServiceNow documents RTO calculations

based on impact tolerability and disruption durations, with category results combined through a minimum and a configured fallback when no intolerable rating occurs [21]. These examples preclude a claim that configurable temporal BIA calculation is new in itself.

The analytical issue is the meaning of each rule under valid and exceptional inputs. Selecting the earliest threshold breach differs from selecting a rating according to its code order. Taking a maximum across dimensions differs from averaging their ratings. Choosing a default duration when no threshold is reached differs from reporting that tolerability has not been established within the horizon. Each procedure can be deterministic, yet yield a different interpretation. Reproducibility consequently requires more than documenting a formula for the normal case.

A complete specification must also address equality at a threshold, missing categories, irregularly spaced periods, invalid configuration, and tied priorities. Where two services share the same decision-relevant attributes, preserving an equal priority may be more defensible than inventing a substantive distinction. If a stable display order is required, the display tie-breaker should not be mistaken for an additional continuity judgment. These requirements arise from the formalization of decision rules; they are not evidence that existing products necessarily lack corresponding safeguards.

Configurability introduces a governance problem as well as a technical capability. The same assessment can produce different outputs under different approved configurations. A traceable record must therefore connect the result to both the assessment and the rule version. It should also distinguish a recalculated result from a historical approval. NIST's emphasis on recording BIA values and their rationale supports this broader view of traceability [10]. For Masar, the relevant research opportunity is to make this chain explicit and testable within the proposed service-prioritization procedure.

2.3.3 Priority Rankings and Recovery Schedules

A priority ranking identifies relative importance or urgency. A recovery schedule assigns actions, resources, and start or completion times while respecting operational constraints. NIST's contingency-planning guidance already recognizes technical

prerequisites in restoration [8]. A high-priority application may depend on an infrastructure service that must be restored first. Recovering the infrastructure first does not mean that its independent business impact exceeds the application's; it means that restoration is constrained by dependencies.

Sahebjamnia et al. formulate integrated continuity and disaster recovery planning as a multi-objective mixed-integer programming problem. BIA supplies tolerability and minimum operating requirements, while the model considers resource capacities, budgets, operating levels, and recovery over discrete periods [6]. Their manufacturing case demonstrates trade-offs between maintaining capability and restoring operations. The model addresses how to allocate resources under specified assumptions, rather than merely deriving service urgency from impact ratings.

El-Shekeil et al.'s PRECESION model examines progressive restoration in enterprise data centers. It represents user requests, services, servers, their dependencies, and limited human recovery capacity. A metaheuristic searches for restoration sequences that improve weighted availability over the recovery horizon [22]. Importance weights are inputs to that scheduling problem. The model thus provides a concrete example of why deriving business priority and executing a feasible restoration sequence are distinct analytical stages.

The distinction also limits comparative claims. An optimization model may produce a better schedule according to its objective function while relying on priorities that were determined elsewhere. A rule-based BIA method may improve consistency in those priorities without optimizing the subsequent schedule. Neither result demonstrates superiority over the other for all continuity decisions. TDRP's outputs can inform planning, but they should not be presented as instructions that automatically account for repair duration, staffing, technical precedence, or changing incident conditions.

2.4 Dependency Modeling and Supporting-Resource Criticality

2.4.1 Service and Resource Relationships

Services are delivered through combinations of activities, information, technology, facilities, personnel, and external providers. NIST's BIA process connects

business processes to required resources, and the Saudi frameworks include dependencies within continuity assessment [1], [8], [15]. A service-first model records these relationships so that resource importance can be explained through supported outcomes. This is especially important for shared resources whose significance is not apparent from their individual technical descriptions.

A dependency statement needs an unambiguous direction and meaning. A relationship indicating that service A requires resource B is different from one indicating that A merely exchanges information with B. A required resource may have substitutes, a manual workaround, or a minimum capacity below normal operating capacity. Consequently, recording an association is an initial representation of dependence, not a complete causal model of service failure. The interpretation adopted by a DSS should remain proportionate to the information actually collected.

PRECESION illustrates a more detailed representation through dependencies between user requests, services, and servers, including relationships within layers [22]. That structure is suitable for restoration planning because the model needs to determine when services become available as servers recover. A simpler service-resource map can still support criticality explanation, but cannot reproduce the same scheduling behavior unless it also contains the required availability conditions and recovery constraints.

Shared dependencies create an additional aggregation issue. If several services rely on one resource, its importance is related to all of them, not just to the first service entered into the system. Conversely, counting the number of supported services alone may undervalue a resource that enables one exceptionally time-sensitive service. The appropriate rule depends on the decision purpose: identifying the strongest inherited requirement, measuring aggregate exposure, and allocating recovery resources are different tasks. A single score should not be assumed to answer all three.

2.4.2 Interdependencies and Impact Propagation

Rinaldi et al. distinguish physical, cyber, geographic, and logical interdependencies among critical infrastructures [23]. Their conceptual framework extends dependency analysis beyond direct technical links. Two services can share exposure because they

occupy the same location, depend on the same external network, or are connected through organizational rules. These relationships help explain how disruption can propagate or affect apparently separate systems simultaneously.

The framework is useful for structuring inquiry, but it does not automatically generate a quantitative propagation model. Such a model requires information about strength, timing, operating states, and possible substitutions. Geographic co-location, for example, indicates common exposure without proving that the failure of one service causes the failure of another. This distinction matters when dependency diagrams are interpreted on dashboards: connectivity is evidence of a relationship, not necessarily a probability or magnitude of cascading loss.

Raptaki et al. offer an empirical approach to deriving process relationships from event logs. Their method discovers a process graph, assigns transition weights, and propagates a selected incident's relative impact through that graph. A Java application uses process-mining components and exports rankings and graph data. Evaluation uses procurement records from one international transportation company together with organizational assessments and interviews [19]. This approach can reveal differences between documented processes and enacted activity.

Its strengths also define its limits. Logged activity supports the observed process model; unlogged work and relationships outside the selected process may remain invisible. Propagated impact values depend on the chosen incident point, graph, and weighting procedure. They are not automatically failure probabilities, service MTPD values, or absolute business-loss estimates. The authors' scenario-sensitive rankings therefore address a different question from TDRP's interpretation of approved service-level temporal assessments [19].

Together, these studies suggest a progression from descriptive dependency inventories to analytical propagation and scheduling models. The appropriate level depends on the research question and available evidence. Masar's stated scope supports linking services with enabling dependencies and explaining inherited criticality. Extending it to infer cascading incidents would require additional modeling and validation beyond what a dependency map alone can establish.

2.4.3 Assigning Criticality to Shared Dependencies

NIST IR 8179 analyses criticality across programs, systems, and components, with attention to relationships among levels and shared components [24]. NIST IR 8286D similarly derives the significance of assets from their contribution to enterprise objectives [10]. These sources establish a clear precedent for business-derived resource criticality. The proposed use of service assessments to explain dependency importance should therefore be positioned as a specific operationalization of an existing principle.

The highest-supported-service rule proposed in the introduction has a comprehensible purpose: a shared dependency should retain the strongest criticality class among the services it supports. This avoids averaging away a highly critical service's requirement. However, this rule is a project policy to be evaluated, not a universal conclusion of the cited guidance. It captures the strongest inherited class but does not distinguish a resource supporting one critical service from another supporting many equally critical services.

The rule also presupposes a common ordering of criticality classes. Numerical codes are insufficient unless their ordering is explicit: in some scales, a smaller code indicates greater criticality. Missing service assessments should not be converted into a low class, and a dependency with no recorded services should not be interpreted as demonstrably unimportant. These cases show why implementation correctness must be evaluated against the semantic definition of the rule rather than against a convenient database maximum operation.

Criticality inheritance and recovery-time assignment should remain separate. A dependency may support several services with different RTOs, and its restoration may need to precede service resumption by a preparation interval. Inheriting the highest class does not determine that interval or establish a feasible dependency recovery target. The literature on contingency planning and scheduling supports maintaining this distinction [8], [22]. For TDRP, inherited criticality is an explanatory output connecting resources to business requirements, not a substitute for operational recovery engineering.

2.5 Decision Support Systems and Comparable Applications

2.5.1 Decision Support Foundations

Decision support systems organize information and analytical capabilities to assist judgment. Power's historical account describes several streams of DSS development, while Arnott et al.'s analysis of DSS research identifies a broad range of systems and methods [25], [26]. A continuity DSS can combine structured records, analytical rules, and interactive presentation without becoming an autonomous decision maker. Its value depends on the connection between those capabilities and the decisions of identifiable users.

Arnott et al. also raise concerns about professional relevance and insufficient identification of clients and users in the research they examined [26]. Their sample covers 1990–2003 and should not be treated as a description of all contemporary DSS research. Nevertheless, the methodological implication remains pertinent: an artifact should state who uses it, what decision it supports, and how improvement is assessed. For Masar, continuity practitioners maintain and interpret service assessments, while authorized decision makers retain responsibility for operational priorities and recovery actions.

Three functions follow from this purpose. The data function preserves services, assessments, configurations, and dependencies. The analytical function applies declared rules consistently. The presentation function exposes results and their rationale in a form practitioners can examine. These are complementary responsibilities. A dashboard that displays a priority without showing its assessment basis may improve visibility while leaving the decision opaque; a correct rule engine without usable explanations may be difficult to challenge or apply appropriately.

Transparency also requires separating recorded evidence from derived results. Users should be able to distinguish an entered impact rating, an approved threshold, a calculated MTPD, and an assigned priority. NIST's BIA register provides a related emphasis on recording values and rationale for communication across organizational levels [10]. The information-system contribution lies partly in preserving these relationships through revision, so that a result remains interpretable after services, assessments, or policies change.

2.5.2 Research Prototypes and Analytical Applications

Research applications demonstrate different forms of continuity-related support. QuantTM presents a browser-based approach to quantifying cyber-threat consequences through business impact and recovery stages [18]. Its output is expected financial loss under threat assumptions. This can support risk discussion, but it differs from assigning service priority from an approved temporal impact matrix. The work is a preprint, and its evidence should be weighed accordingly rather than treated as equivalent to independently replicated operational results.

Raptaki et al.'s application supports process discovery and impact propagation, connecting event data to scenario-sensitive activity rankings [19]. Its implementation is important because it demonstrates how analytical models can be integrated with practical data processing and export. However, the quality of its output depends on event-log coverage and interpretation. Automation can reduce some manual modelling effort while leaving substantive organizational judgments unresolved.

Russo et al.'s self-assessment tool addresses BCM preparedness through structured questions, weighted scoring, explanatory guidance, and visual results [27]. The study includes practitioner engagement and organizational interviews, providing evidence about interpretation and perceived usefulness. Its decision target is organizational improvement, not temporal service criticality. A preparedness score cannot be relabeled as a recovery priority, and anticipated recovery benefits should not be reported as measured incident outcomes.

2.5.3 Continuity Management Software

Commercial and open-source-oriented continuity tools provide essential context because several proposed functions already exist in practice. The comparison in Table 2.3 uses provider documentation and distinguishes disclosed behavior from unreported detail. No controlled product trial underlies the comparison. Accordingly, an unspecified capability is recorded as unknown within the reviewed material, rather than treated as absent from the product.

Table 2.3: Comparable continuity tools: documented functionality and limits of the reviewed evidence.

System	Documented functions relevant to this thesis	Comparison boundary
ServiceNow [21], [28]	BIA workflow; configured tolerability and disruption durations; RTO category calculations and minimum aggregation; recovery-tier mappings.	Strong precedent for configured calculation. Reviewed documentation does not validate equivalence to TDRP or comparative recovery effectiveness.
GlobalSuite [20]	Service-level temporal impact table; manual or automatic MTPD; configurable calculation and assessment-consolidation criteria.	Close temporal BIA precedent. Exceptional-input handling, full ranking semantics, and dependency inheritance are not established by the reviewed page.
GRASP GRC [29]	Time-horizon impact assessment; recovery objectives; linked assets, providers, and process chains; recorded rationale.	Documents an assessment workflow. Complete automatic derivation and aggregation algorithms are not disclosed in the reviewed material.
verinice [30]	BCM domain; business-process assessment, BIA, and dependency evaluation; integration with other management domains.	Relevant alternative tool family. Precise temporal calculations and licensing across product generations are not established here.
Fusion [31]	Integrated continuity management, BIA, and dependency-related planning capabilities.	Provider-level description; detailed calculation rules and independent comparative results are not established.
Riskonnect [32]	Continuity management, business impact assessment, and dependency-related planning capabilities.	Provider-level description; detailed calculation rules and independent comparative results are not established.

ServiceNow's BIA documentation describes structured assessment workflows, while its recovery-objective documentation specifies calculations using tolerability, disruption durations, and configured mappings [21], [28]. The broader solution brief places BIA within continuity planning and associated management functions [33]. This is a close functional comparison for Masar because it demonstrates that calculation and workflow are already combined. The relevant distinction is the exact semantics and evidence available for the proposed method, not the existence of a web interface for BIA.

GlobalSuite is particularly relevant to service-level temporal assessment. Its Pre-BIA documentation associates a service with an impact table over configured durations, permits manual or automatic MTPD, and provides consolidation rules across assessments [20]. The availability of maximum, minimum, and averaging options shows that configuration can change the interpretation of the assessment. Its documented temporal calculation is substantive prior art. The reviewed page does not fully specify exceptional-input behavior or an end-to-end evaluation of service ranking and dependency inheritance; these remain limits of the available evidence.

GRASP GRC documents impact assessment across time horizons, recovery objectives, linked assets and providers, and the recording of rationale [29]. Its assessment view connects temporal analysis to dependencies and continuity planning. The reviewed material does not expose a complete automatic derivation algorithm for every recovery parameter. This supports a distinction between a documented assessment workflow and a reproducible computational specification, without implying that practitioners receive no useful support from the workflow.

The reviewed verinice BCM page describes BIA, process and dependency evaluation, and integration with broader management domains [30]. It provides a further example beyond the largest commercial suites, but product generation and licensing claims require care. The documentation's discussion of open-source development should not be extended automatically to every current hosted offering. For the present comparison, the relevant evidence is the disclosed BCM and dependency functionality; exact temporal derivation and shared-dependency aggregation remain unverified.

Fusion and Riskonnect describe broader continuity-management capabilities, including BIA and dependency-related planning [31], [32]. Their product descriptions establish the availability of integrated continuity platforms, but provide less detail about the exact calculations relevant to this thesis than the ServiceNow and GlobalSuite documentation. Marketing statements about resilience or efficiency are not substitutes for independent evaluations. Conversely, limited public algorithmic disclosure is not evidence of defective implementation.

The software comparison therefore changes the formulation of the research gap. Masar cannot be justified simply by claiming that existing tools do not support BIA, time-dependent assessment, dependencies, or dashboards. The defensible objective is to specify and evaluate a bounded, transparent method whose assessment inputs, rule configuration, recovery outputs, relative priorities, and dependency classifications can be examined together. Whether this offers practical advantages over existing tools requires comparative evidence and cannot be assumed from the prototype's proposed functionality.

2.6 Research Methods and Evaluation Evidence

2.6.1 Design Science and Artifact Development

Design-science research is appropriate when the research contribution includes a purposeful artifact addressing an organizational problem. Hevner et al. emphasize the relationship among relevance, rigor, artifact construction, and evaluation [3]. Peffers et al. articulate activities covering problem identification, solution objectives, design and development, demonstration, evaluation, and communication [4]. These perspectives explain why developing Masar is necessary but not sufficient: implementation must be connected to an explicit research problem and supported by evidence about the resulting artifact.

TDRP and Masar are analytically distinguishable artifacts. TDRP specifies the decision procedure, while Masar realizes it in software. A defect in a threshold rule is different from a programming error in executing a correct rule. Similarly, an accurate computation may still be presented in a way that users misunderstand. Separating these layers makes it possible to evaluate methodological consistency, implementation correctness, and practical interpretation without treating them as one undifferentiated claim of system quality.

The related studies use several development strategies. Sahebjamnia et al. formalize a planning model and examine its application in a manufacturing setting [6]. Raptaki et al. combine analytical development with process data and organizational feedback [19]. Russo et al. develop and refine a self-assessment artifact through expert and organizational engagement [27]. These approaches offer complementary

precedents, but none prescribes the exact evaluation design for TDRP. The evaluation must follow the properties the thesis claims to establish.

2.6.2 Correctness, Usefulness, and Operational Effectiveness

Venable et al.'s Framework for Evaluation in Design Science (FEDS) distinguishes formative from summative evaluation and artificial from naturalistic evaluation settings [34]. Formative evaluation informs improvement; summative evaluation judges the artifact against expectations. Artificial settings permit controlled examination of selected properties, whereas naturalistic settings examine use in real organizational contexts. These dimensions should not be reduced to a simple sequence in which every early test is formative and every later interview is proof of effectiveness.

For a deterministic method, correctness can be examined through independently specified cases with known expected outcomes. Relevant cases include threshold equality, an impact already unacceptable at the first period, no observed threshold crossing, incomplete assessments, tied priorities, and several services sharing a dependency. Expected results must come from the declared semantics rather than being copied from the implementation being tested. Otherwise, the same error may appear in both calculation and reference output. These are implications of the artifact's claims, not evaluations already completed in the present thesis.

Consistency concerns how results behave across repeated execution and controlled input changes. Repeating an assessment under the same configuration should reproduce its result. Reordering stored records should not alter a priority unless order is explicitly part of the policy. If higher severity is intended never to reduce criticality, that property should be checked over valid configurations. Such tests assess the internal consequences of the chosen method; they do not establish that the organization's original thresholds reflect actual harm accurately.

Sensitivity analysis addresses a different question: how strongly conclusions depend on plausible changes in assumptions. The asset-criticality study uses sensitivity analysis alongside expert face validation [9]. For TDRP, changing approved thresholds or time grids could alter classifications and rankings even when the implementation is correct. Reporting these changes would help distinguish robust conclusions from

priorities that depend heavily on a particular configuration. Stability is not always desirable, however: an appropriately responsive method should change its output when a materially different tolerability judgment is introduced.

Usefulness and interpretability require human evidence. Practitioners may be asked to explain a displayed priority, identify its decisive impact, trace a shared dependency's class, or recognize an incomplete assessment. Successful completion supports claims about particular tasks and users. General satisfaction alone is weaker evidence that the decision rationale was understood. The practitioner-oriented evaluation in the self-assessment study and the organizational feedback in the process-mining study illustrate the value of contextual engagement, while retaining the limits of their respective samples [19], [27].

Operational effectiveness requires stronger evidence still. A prototype demonstration cannot establish reduced outage duration, lower incident losses, or improved organizational resilience. Those outcomes depend on staffing, infrastructure, procedures, disruption characteristics, and actual use of the system. The literature reviewed here frequently combines cases, scenarios, expert assessment, and sensitivity analysis. It supports feasibility and contextual usefulness more directly than universal causal claims about recovery performance. The evaluation of Masar should preserve this distinction when interpreting its own results.

2.6.3 Transferability and Evidence Quality

Single-case studies provide detailed context but limited grounds for generalization across sectors. The AMM, container-terminal, gas-plant, manufacturing, and procurement studies address materially different operating environments [6], [9], [16], [17], [19]. Their diversity supports the general relevance of continuity analysis, yet does not establish that one shared numerical scale is valid across them. Transferability should concern the reasoning structure and adaptation requirements, with local parameters justified separately.

For this thesis, the literature supports a proportionate evaluation: explicit rule specification, independently checked cases, tests of configuration behavior and shared dependencies, and practitioner examination of traceability and interpretation. Claims

about organizational benefits should be limited to the evidence actually obtained. This provides a coherent route from the introductory objectives to a design-science contribution without requiring a prototype study to demonstrate effects that only deployment and longitudinal observation could establish.

2.7 Comparative Synthesis and Research Gap

The reviewed literature establishes a connected chain of decisions. BCM defines the organizational need to sustain prioritized outcomes. BIA identifies the consequences of interruption and their temporal significance. Prioritization methods translate judgments into classes or relative rankings. Dependency models connect those outcomes to enabling resources, while planning and scheduling methods consider how restoration can be executed. DSS applications integrate selected parts of this chain for practitioner use. Table 2.4 summarizes the implications for the proposed research.

Table 2.4: Synthesis of prior contributions and implications for TDRP and Masar.

Decision area	Established contribution	Implication for the proposed research
Continuity requirements	Contextual BIA, recovery objectives, and resources in government guidance [8], [15].	Preserve the organizational meaning of parameters; do not claim that software alone demonstrates compliance.
Temporal assessment	Time-indexed impact matrices in AMM BIA and configurable temporal calculations in product documentation [16], [20], [21].	Specify the exact rule semantics, boundaries, missing-data behavior, and configuration used by TDRP.
Criticality aggregation	Weighted, fuzzy, and probabilistic assessment of business-derived asset importance [9].	Explain why a chosen aggregation policy fits the decision; do not assume a simple rule is universally superior.
Recovery execution	Resource-constrained operating trajectories and dependency-aware restoration scheduling [6], [22].	Keep service ranking distinct from operational sequencing and capacity allocation.
Supporting dependencies	Business-derived asset criticality and process-level impact propagation [10], [19], [24].	Define shared-dependency inheritance explicitly; distinguish classification from causal propagation.

Decision area	Established contribution	Implication for the proposed research
Artifact evaluation	Design-science methods and evaluation strategies separating properties and settings [3], [4], [34].	Evaluate method, implementation, and practitioner interpretation separately; bound operational claims.

Several capabilities are already established. Standards and government guidance support contextual BIA, recovery requirements, and dependency identification. Research demonstrates temporal impact matrices, business-derived asset criticality, mathematical recovery planning, and process-based propagation. Product documentation describes configurable temporal calculations and integrated continuity workflows [6], [8], [9], [16], [19], [20], [21]. A broad claim that prior work lacks time sensitivity, decision rules, or service-resource relationships would therefore be unsupported.

The remaining research problem is more specific. The thesis seeks a fully stated, configurable procedure that connects service-level temporal assessments to criticality, MTPD, RTO, and relative priorities, then explains the criticality assigned to supporting dependencies. The reviewed sources provide important parts of this procedure, but do not collectively validate the exact combination and interpretation proposed for TDRP. Its contribution must be demonstrated through the coherence of that specification, its treatment of exceptional inputs, and the evaluation of its implementation and explanations.

This gap is an artifact-specific design and evaluation problem, not proof of an unoccupied software market. GlobalSuite and ServiceNow are particularly close prior art for temporal rule execution. NIST supplies prior art for business-derived criticality, and the research on scheduling establishes a clear boundary on what a ranking can claim. These precedents narrow the novelty claim while improving the basis for design. They provide concepts and comparison points against which TDRP can be examined rather than reasons to disregard the proposed project.

Three requirements emerge from the synthesis. First, temporal interpretation must be explicit: ratings, time horizons, threshold semantics, and recovery mappings must retain distinct meanings. Second, the chain from input to output must be

traceable, including configuration and shared-dependency rules. Third, evaluation must distinguish correct execution from appropriate policy and practical usefulness. These requirements align with the introduction's emphasis on correctness, consistency, and traceability, while leaving real-incident effectiveness outside what a prototype alone can establish.

Masar operationalizes this bounded contribution as a web-based DSS. Its intended dashboard presents approved assessment results and their context; it does not infer recovery procedures or execute operational decisions. The methodology chapter can consequently build on a precise literature-derived foundation: formalize the proposed rules, justify their assumptions, describe their implementation, and establish evaluation criteria corresponding to each claimed property.

2.8 Chapter Summary

The literature identifies BIA as a central connection between organizational continuity objectives and recovery requirements. Time-dependent assessment distinguishes services by the onset of unacceptable consequences, while criticality and dependency analysis explain why enabling resources matter. Research offers matrix-based, multi-criteria, probabilistic, process-mining, and optimization approaches, each with different inputs and decision outputs. Comparable software already supports important elements of temporal assessment, configured calculations, and continuity workflows.

The review supports a bounded contribution for TDRP: an explicit and evaluable procedure linking service assessments, recovery requirements, relative priorities, and supporting-dependency criticality. It also establishes the limits of that contribution. A reproducible priority is not a guaranteed appropriate policy, a dependency class is not a restoration schedule, and a working prototype is not evidence of improved recovery during actual disruptions. These distinctions provide the basis for the design and evaluation of Masar in the next chapter.

Chapter 3: Methodology/ Research Design

This chapter of the proposal should outline the design and methodology of your research. The basis for the choice of research method should be whether it will help you answer your research question(s).

Chapter 4: Results and Discussions

Note: Results and Discussions can be presented in a single chapter or in two different chapters depending on the length of your thesis.

Chapter 5: Analysis/ Discussions

Note: Similarly, Results and Discussions can be presented in a single chapter or in two different chapters depending on the length of your thesis.

Chapter 6: Conclusions

This chapter contains conclusions, limitations, and recommendations – so what is the theory?

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Appendices

Appendix A

Title

Start each appendix on a new page. Place appendices in the same order as they are referred to in the body of the thesis. That is, the first appendix referred to should be Appendix A, the second appendix referred to should be Appendix B, and so on. Appendix formatting can be different to the main document.

Appendix B

List of Publications

Add the first page of each research papers, journal articles, and conference proceedings that have arisen from the research being presented in your thesis.